

## PVTRACE · DESIGN DECISION RECORD

# PVTrace recommendations accepted

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## DOCUMENT CONTROL

|             |                                                               |         |                |
|-------------|---------------------------------------------------------------|---------|----------------|
| Document ID | PVT-DDR-002                                                   | Version | 0.1            |
| Status      | <b>Draft</b>                                                  | Date    | 2026-09-25     |
| Author      | Amish Chadha                                                  | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                                 |         |                |
| Repository  | github.com/BoujeeEnjinia1701/pvtrace                          |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                            |
|---------|------------|--------------|-----------------------------------------------------------------------------------|
| 0.1     | 2026-09-25 | Amish Chadha | Record Amish's acceptance of all recommendations and the changes made in the repo |

# 0002: Recommendations accepted

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- **Date:** 2026-09-25
- **Status:** accepted. Every item with a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items with no recommendation stay proposed, awaiting Amish.

## ■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." This record lists every PVTrace item that the decision closes, what changed in the repo because of it, and what stays open. The items come from the TRL 2 review note ( docs/REVIEW.md , /populate session) and PVT-DDR-001. Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, so nothing here is built, bought or tested.

## ■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation

| #                  | Item                                        | Decision                                                                                                                                | What changed in the repo                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                  | Load type                                   | Capacitive load                                                                                                                         | Status wording only; already in PVT-PRC-001                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2                  | Rating                                      | 100 V, 20 A, below the 120 V DC extra-low-voltage limit                                                                                 | Status wording only; R1, R2 unchanged                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3                  | Load capacitance                            | 6.6 mF, three 2200 µF 160 V capacitors                                                                                                  | Status wording only; already in <code>bom/bom.csv</code> line 5 and the model                                                                                                                                                                                                                                                                                                                                                                             |
| 4                  | Isolation                                   | Option B: digital isolator and isolated DC-DC converter                                                                                 | Status wording only; already in <code>bom/bom.csv</code> line 14 and the model                                                                                                                                                                                                                                                                                                                                                                            |
| 5                  | Display                                     | 2.8 in TFT plus phone export                                                                                                            | Status wording only                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6                  | Irradiance reference                        | Reference cell calibrated once against a pyranometer                                                                                    | Status wording only; the calibration itself is TRL 4 work, on hold                                                                                                                                                                                                                                                                                                                                                                                        |
| 7                  | Controller                                  | ESP32 display board                                                                                                                     | Status wording only                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8                  | Grade thresholds                            | A 90 % or more of nameplate, B 80 to 90 %, C 70 to 80 %, reject below 70 % or with a visible safety defect, in a partner-editable table | R10 wording in PVT-REQ-001 v0.4 and PVT-CAL-001 v0.2                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9                  | Budget                                      | Raise <code>budget_usd</code> from \$150 to \$165                                                                                       | <code>project.yaml</code> <code>budget_usd</code> 150 to 165; README budget line; R17 in PVT-REQ-001 v0.4 changes from not met (\$163 against \$150) to met (\$164 against \$165, \$1 margin); PVT-PRB-001 v0.4 constraint                                                                                                                                                                                                                                |
| TRL 2 note item 11 | <code>pitch</code> and <code>problem</code> | No change, as recommended                                                                                                               | None                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 12                 | Heat and sun (R13)                          | Light grey case, charger with temperature cut-off and a display sun hood                                                                | New BOM line 15, printed PETG sun hood, 86 x 64 x 18 mm, \$1.00; hood added to <code>cad/src/model.py</code> (no clashes; top at Z 98, level with the isolator knob, so the R12 envelope is unchanged); STEP and STL re-exported; PVT-DWG-001 Rev P1 to P2; concept media refreshed with callout 15; PVT-CAL-001 v0.2: parts \$163 to \$164, mass 1.41 kg to 1.43 kg. R13 display readability stays not met on paper until a field check (TRL 4, on hold) |
| 13                 | Case size                                   | 220 x 130 x 80 mm                                                                                                                       | Status wording only; already in the model and drawing                                                                                                                                                                                                                                                                                                                                                                                                     |

## Items still open

Table 2. Proposed, awaiting Amish

| #  | Item                                                                                                                                            | Why it stays open                                 |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 10 | First co-design partner: a second-life panel refurbisher or recycler, or a TVET solar course                                                    | No preference stated; no recommendation to accept |
| 11 | R3 at worst-case capacitor tolerance (16.4 ms at -20 %): a fourth capacitor, selected capacitors of 6.4 mF or more, or relaxing the 20 ms bound | No recommendation was made at TRL 3               |

### Cross-repo actions

None. CalRig's 10 to 40 °C range limit for the temperature probe (PVT-CAL-001 section 4) was a note, not a recommendation to change CalRig.

### On hold (TRL 4)

Reference cell calibration, the display readability field check, any purchase, build or bench test. These follow from the decisions above but are TRL 4 work and stay on hold by Amish's instruction.

## ■ Consequences

- The budget is \$165 and the priced BOM is \$164, so R17 is met with a \$1 margin. Any later addition, such as a fourth capacitor for item 11 (about +\$6), would break it again.
- Requirements now stand at ten met, three not met (R3 at tolerance, R13 display readability, R16 out of scope), two at risk (R7, R8) and two not verifiable at TRL 3 (R6, R9).
- The sun hood adds about 15 g; mass is 1.43 kg against 1.5 kg (R12).
- `trl` and `trl_target` stay at 3.